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MySQL Practical Assignment – Employee Data Analysis

DATABASE SETUP:

```
CREATE DATABASE JYO;

USE JYO;
```

This database is created to perform employee data analysis using different SQL concepts such as constraints, filtering, sorting, aggregate functions, grouping, HAVING clauses.

Stage 1: Create the Employees Table

Query 1:

```
CREATE TABLE employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(100) NOT NULL,
    email VARCHAR(100) NOT NULL UNIQUE,
    department VARCHAR(50) NOT NULL,
    job_role VARCHAR(50) NOT NULL,
    salary DECIMAL(10,2) CHECK (salary>0),
    age INT CHECK(age>=18),
    city VARCHAR(50) DEFAULT 'Vijayawada'
);
```

Query 2: DESC employees;

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
emp_id	int	NO	PRI	NULL	
emp_name	varchar(100)	NO		NULL	
email	varchar(100)	NO	UNI	NULL	
department	varchar(50)	NO		NULL	
job_role	varchar(50)	NO		NULL	
salary	decimal(10,2)	YES		NULL	
age	int	YES		NULL	
city	varchar(50)	YES		Vijayawada	

Stage 2: Insert Employee Data

To insert 20 realistic employee records with different departments, cities, salaries, ages, and job roles. Use all 20 records in a single section.

INSERT INTO employees (emp_id, emp_name, email, department, job_role, salary, age, city) VALUES

(101, 'Rahul Sharma', 'rahul.sharma@gmail.com', 'IT', 'Software Developer', 60000, 24, 'Hyderabad'),

(102, 'Priya Reddy', 'priya.reddy@gmail.com', 'HR', 'HR Executive', 45000, 26, 'Vijayawada'),

(103, 'Arjun Kumar', 'arjun.kumar@gmail.com', 'Finance', 'Financial Analyst', 55000, 28, 'Bangalore'),

(104, 'Sneha Patel', 'sneha.patel@gmail.com', 'Marketing', 'Marketing Executive', 50000, 25, 'Chennai'),

(105, 'Vikram Singh', 'vikram.singh@gmail.com', 'Sales', 'Sales Executive', 45000, 27, 'Hyderabad'),

(106, 'Anjali Verma', 'anjali.verma@gmail.com', 'IT', 'Software Tester', 55000, 23, 'Pune'),

(107, 'Karthik Rao', 'karthik.rao@gmail.com', 'Finance', 'Accountant', 50000, 30, 'Vijayawada'),

(108, 'Divya Nair', 'divya.nair@gmail.com', 'HR', 'Recruiter', 45000, 29, 'Bangalore'),

(109, 'Rohan Mehta', 'rohan.mehta@gmail.com', 'Marketing', 'Content Strategist', 60000, 26, 'Chennai'),

(110, 'Meera Iyer', 'meera.iyer@gmail.com', 'Sales', 'Business Development Executive', 55000, 24, 'Pune'),

(111, 'Suresh Babu', 'suresh.babu@gmail.com', 'IT', 'Data Analyst', 65000, 31, 'Hyderabad'),

(112, 'Kavya Singh', 'kavya.singh@gmail.com', 'HR', 'HR Manager', 70000, 35, 'Vijayawada'),

(113, 'Aman Gupta', 'aman.gupta@gmail.com', 'Finance', 'Senior Accountant', 65000, 33, 'Bangalore'),

(114, 'Pooja Das', 'pooja.das@gmail.com', 'Marketing', 'SEO Specialist', 50000, 27, 'Chennai'),

(115, 'Naveen Kumar', 'naveen.kumar@gmail.com', 'Sales', 'Sales Manager', 70000, 34, 'Pune'),

(116, 'Lakshmi Devi', 'lakshmi.devi@gmail.com', 'IT', 'Project Manager', 80000, 36, 'Hyderabad'),

(117, 'Manoj Reddy', 'manoj.reddy@gmail.com', 'Finance', 'Finance Manager', 80000, 38, 'Vijayawada'),

(118, 'Swathi Krishna', 'swathi.krishna@gmail.com', 'Marketing', 'Graphic Designer', 55000, 25, 'Bangalore'),

(119, 'Ajay Varma', 'ajay.varma@gmail.com', 'Sales', 'Sales Associate', 40000, 22, 'Chennai'),

(120, 'Neha Joshi', 'neha.joshi@gmail.com', 'IT', 'UI UX Designer', 60000, 28, NULL);

Stage 3: SELECT Queries

Query 1: Display all employee records

This query is used to display all the details of employees from the employees table.

SELECT *FROM employees;

Output:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	email	department	job_role	salary	age	city
101	Rahul Sharma	rahul.sharma@gmail.com	IT	Software Developer	60000.00	24	Hyderabad
102	Priya Reddy	priya.reddy@gmail.com	HR	HR Executive	45000.00	26	Vijayawada
103	Arjun Kumar	arjun.kumar@gmail.com	Finance	Financial Analyst	55000.00	28	Bangalore
104	Sneha Patel	sneha.patel@gmail.com	Marketing	Marketing Executive	50000.00	25	Chennai
105	Vikram Singh	vikram.singh@gmail.com	Sales	Sales Executive	45000.00	27	Hyderabad
106	Anjali Verma	anjali.verma@gmail.com	IT	Software Tester	55000.00	23	Pune
107	Karthik Rao	karthik.rao@gmail.com	Finance	Accountant	50000.00	30	Vijayawada
108	Divya Nair	divya.nair@gmail.com	HR	Recruiter	45000.00	29	Bangalore
109	Rohan Mehta	rohan.mehta@gmail.com	Marketing	Content Strategist	60000.00	26	Chennai
110	Meera Iyer	meera.iyer@gmail.com	Sales	Business Development Executive	55000.00	24	Pune
111	Suresh Babu	suresh.babu@gmail.com	IT	Data Analyst	65000.00	31	Hyderabad
112	Kavya Singh	kavya.singh@gmail.com	HR	HR Manager	70000.00	35	Vijayawada
113	Aman Gupta	aman.gupta@gmail.com	Finance	Senior Accountant	65000.00	33	Bangalore
114	Pooja Das	pooja.das@gmail.com	Marketing	SEO Specialist	50000.00	27	Chennai
115	Naveen Kumar	naveen.kumar@gmail.com	Sales	Sales Manager	70000.00	34	Pune
116	Lakshmi Devi	lakshmi.devi@gmail.com	IT	Project Manager	80000.00	36	Hyderabad
117	Manoj Reddy	manoj.reddy@gmail.com	Finance	Finance Manager	80000.00	38	Vijayawada
118	Swathi Krishna	swathi.krishna@gmail.com	Marketing	Graphic Designer	55000.00	25	Bangalore
119	Ajay Varma	ajay.varma@gmail.com	Sales	Sales Associate	40000.00	22	Chennai
120	Neha Joshi	neha.joshi@gmail.com	IT	UI UX Designer	60000.00	28	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Query 2: Display specific columns: Employee Name, Department, Salary, and City

This query is used to display only the required details of employees.

Here, we are displaying employee name, department, salary, and city.

SELECT emp_name, department, salary, city FROM employees;

Output :

Result Grid | Filter Rows: | Export:

emp_name	department	salary	city
Rahul Sharma	IT	60000.00	Hyderabad
Priya Reddy	HR	45000.00	Vijayawada
Arjun Kumar	Finance	55000.00	Bangalore
Sneha Patel	Marketing	50000.00	Chennai
Vikram Singh	Sales	45000.00	Hyderabad
Anjali Verma	IT	55000.00	Pune
Karthik Rao	Finance	50000.00	Vijayawada
Divya Nair	HR	45000.00	Bangalore
Rohan Mehta	Marketing	60000.00	Chennai
Meera Iyer	Sales	55000.00	Pune
Suresh Babu	IT	65000.00	Hyderabad
Kavya Singh	HR	70000.00	Vijayawada
Aman Gupta	Finance	65000.00	Bangalore
Pooja Das	Marketing	50000.00	Chennai
Naveen Kumar	Sales	70000.00	Pune
Lakshmi Devi	IT	80000.00	Hyderabad
Manoj Reddy	Finance	80000.00	Vijayawada
Swathi Krishna	Marketing	55000.00	Bangalore
Ajay Varma	Sales	40000.00	Chennai
Neha Joshi	IT	60000.00	NULL

MySQL Practical Assignment – Employee Data Analysis

1. **Retrieve employees based on specific conditions**

This query is used to find employees who are working in the IT department and whose salary is greater than 50000.

The AND operator is used because both conditions should be satisfied.

SELECT * FROM employees WHERE department = 'IT' AND salary > 50000;

Output:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	email	department	job_role	salary	age	city
101	Rahul Sharma	rahul.sharma@gmail.com	IT	Software Developer	60000.00	24	Hyderabad
106	Anjali Verma	anjali.verma@gmail.com	IT	Software Tester	55000.00	23	Pune
111	Suresh Babu	suresh.babu@gmail.com	IT	Data Analyst	65000.00	31	Hyderabad
116	Lakshmi Devi	lakshmi.devi@gmail.com	IT	Project Manager	80000.00	36	Hyderabad
120	Neha Joshi	neha.joshi@gmail.com	IT	UI UX Designer	60000.00	28	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Stage 4: WHERE and Comparison Operators

1)EQUAL to (=) : Display employees whose salary is exactly 65000.

The equal to operator is used to find employees whose salary exactly matches the given value.

SELECT * FROM employees WHERE salary = 65000;

Output :

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	email	department	job_role	salary	age	city
111	Suresh Babu	suresh.babu@gmail.com	IT	Data Analyst	65000.00	31	Hyderabad
113	Aman Gupta	aman.gupta@gmail.com	Finance	Senior Accountant	65000.00	33	Bangalore
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

2. Greater than(>) : Display employees earning more than 60000.

The greater than operator is used to find values that are higher than the given value.

SELECT * FROM employees WHERE salary > 60000;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	email	department	job_role	salary	age	city
111	Suresh Babu	suresh.babu@gmail.com	IT	Data Analyst	65000.00	31	Hyderabad
112	Kavya Singh	kavya.singh@gmail.com	HR	HR Manager	70000.00	35	Vijayawada
113	Aman Gupta	aman.gupta@gmail.com	Finance	Senior Accountant	65000.00	33	Bangalore
115	Naveen Kumar	naveen.kumar@gmail.com	Sales	Sales Manager	70000.00	34	Pune
116	Lakshmi Devi	lakshmi.devi@gmail.com	IT	Project Manager	80000.00	36	Hyderabad
117	Manoj Reddy	manoj.reddy@gmail.com	Finance	Finance Manager	80000.00	38	Vijayawada
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3.Less than (<) : Display employees whose age is below 25.

The less than operator is used to find values that are lower than the given value.

SELECT * FROM employees WHERE age < 25;

emp_id	emp_name	email	department	job_role	salary	age	city
101	Rahul Sharma	rahul.sharma@gmail.com	IT	Software Developer	60000.00	24	Hyderabad
106	Anjali Verma	anjali.verma@gmail.com	IT	Software Tester	55000.00	23	Pune
110	Meera Iyer	meera.iyer@gmail.com	Sales	Business Development Executive	55000.00	24	Pune
119	Ajay Varma	ajay.varma@gmail.com	Sales	Sales Associate	40000.00	22	Chennai
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

4.Greater than or Equal to (>=) : Display employees earning greater than or equal to 80000.

The greater than or equal to operator is used to find values that are either greater than or exactly equal to the given value.

SELECT * FROM employees WHERE salary >= 80000;

emp_id	emp_name	email	department	job_role	salary	age	city
116	Lakshmi Devi	lakshmi.devi@gmail.com	IT	Project Manager	80000.00	36	Hyderabad
117	Manoj Reddy	manoj.reddy@gmail.com	Finance	Finance Manager	80000.00	38	Vijayawada
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

5. **Less than or Equal to (<=): Display employees whose age is less than or equal to 30.**
   # The less than or equal to operator is used to find values that are either smaller than or exactly equal to the given value.
   SELECT * FROM employees WHERE age <= 30;

emp_id	emp_name	email	department	job_role	salary	age	city
101	Rahul Sharma	rahul.sharma@gmail.com	IT	Software Developer	60000.00	24	Hyderabad
102	Priya Reddy	priya.reddy@gmail.com	HR	HR Executive	45000.00	26	Vijayawada
103	Arjun Kumar	arjun.kumar@gmail.com	Finance	Financial Analyst	55000.00	28	Bangalore
104	Sneha Patel	sneha.patel@gmail.com	Marketing	Marketing Executive	50000.00	25	Chennai
105	Vikram Singh	vikram.singh@gmail.com	Sales	Sales Executive	45000.00	27	Hyderabad
106	Anjali Verma	anjali.verma@gmail.com	IT	Software Tester	55000.00	23	Pune
107	Karthik Rao	karthik.rao@gmail.com	Finance	Accountant	50000.00	30	Vijayawada
108	Divya Nair	divya.nair@gmail.com	HR	Recruiter	45000.00	29	Bangalore
109	Rohan Mehta	rohan.mehta@gmail.com	Marketing	Content Strategist	60000.00	26	Chennai
110	Meera Iyer	meera.iyer@gmail.com	Sales	Business Development Executive	55000.00	24	Pune
114	Pooja Das	pooja.das@gmail.com	Marketing	SEO Specialist	50000.00	27	Chennai
118	Swathi Krishna	swathi.krishna@gmail.com	Marketing	Graphic Designer	55000.00	25	Bangalore
119	Ajay Varma	ajay.varma@gmail.com	Sales	Sales Associate	40000.00	22	Chennai
120	Neha Joshi	neha.joshi@gmail.com	IT	UI UX Designer	60000.00	28	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

6. **Not Equal to (<> & !=): Display employees whose department is not HR.**

The not equal operator is used to exclude a particular value. Here, I am displaying employees who are not working in the HR department.

Both != and <> can be used for not equal in SQL.

SELECT * FROM employees WHERE department != 'HR';

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	email	department	job_role	salary	age	city
102	Priya Reddy	priya.reddy@gmail.com	HR	HR Executive	45000.00	26	Vijayawada
103	Arjun Kumar	arjun.kumar@gmail.com	Finance	Financial Analyst	55000.00	28	Bangalore
104	Sneha Patel	sneha.patel@gmail.com	Marketing	Marketing Executive	50000.00	25	Chennai
105	Vikram Singh	vikram.singh@gmail.com	Sales	Sales Executive	45000.00	27	Hyderabad
107	Karthik Rao	karthik.rao@gmail.com	Finance	Accountant	50000.00	30	Vijayawada
108	Divya Nair	divya.nair@gmail.com	HR	Recruiter	45000.00	29	Bangalore
109	Rohan Mehta	rohan.mehta@gmail.com	Marketing	Content Strategist	60000.00	26	Chennai
110	Meera Iyer	meera.iyer@gmail.com	Sales	Business Development Executive	55000.00	24	Pune
112	Kavya Singh	kavya.singh@gmail.com	HR	HR Manager	70000.00	35	Vijayawada
113	Aman Gupta	aman.gupta@gmail.com	Finance	Senior Accountant	65000.00	33	Bangalore
114	Pooja Das	pooja.das@gmail.com	Marketing	SEO Specialist	50000.00	27	Chennai
115	Naveen Kumar	naveen.kumar@gmail.com	Sales	Sales Manager	70000.00	34	Pune
117	Manoj Reddy	manoj.reddy@gmail.com	Finance	Finance Manager	80000.00	38	Vijayawada
118	Swathi Krishna	swathi.krishna@gmail.com	Marketing	Graphic Designer	55000.00	25	Bangalore
119	Ajay Varma	ajay.varma@gmail.com	Sales	Sales Associate	40000.00	22	Chennai
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

In this stage, I learned how to use the WHERE clause with different comparison operators to filter employee records. These operators help me retrieve records based on conditions such as equal to, greater than, less than, and not equal to.

Stage 5: Logical Operators

1. AND Operator : Find employees from the IT department AND earning above 60,000.

The AND operator is used when both conditions should be true. I am finding employees who are from the IT department

SELECT * FROM employees WHERE department = 'IT' AND salary > 60000;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	email	department	job_role	salary	age	city
111	Suresh Babu	suresh.babu@gmail.com	IT	Data Analyst	65000.00	31	Hyderabad
116	Lakshmi Devi	lakshmi.devi@gmail.com	IT	Project Manager	80000.00	36	Hyderabad
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

2. OR Operator : Find employees who belong to either the IT department OR HR department.

The OR operator is used when at least one of the given conditions should be true

SELECT * FROM employees WHERE department = 'IT' OR department = 'HR';

emp_id	emp_name	email	department	job_role	salary	age	city
101	Rahul Sharma	rahul.sharma@gmail.com	IT	Software Developer	60000.00	24	Hyderabad
102	Priya Reddy	priya.reddy@gmail.com	HR	HR Executive	45000.00	26	Vijayawada
106	An Anjali Verma	ıjali.verma@gmail.com	IT	Software Tester	55000.00	23	Pune
108	Divya Nair	divya.nair@gmail.com	HR	Recruiter	45000.00	29	Bangalore
111	Suresh Babu	suresh.babu@gmail.com	IT	Data Analyst	65000.00	31	Hyderabad
112	Kavya Singh	kavya.singh@gmail.com	HR	HR Manager	70000.00	35	Vijayawada
116	Lakshmi Devi	lakshmi.devi@gmail.com	IT	Project Manager	80000.00	36	Hyderabad
120	Neha Joshi	neha.joshi@gmail.com	IT	UI UX Designer	60000.00	28	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. **NOT Operator** : Find employees who are NOT from the IT department.

   # The NOT operator is used to exclude records that satisfy a condition.

   SELECT * FROM employees WHERE NOT department = 'IT';

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	email	department	job_role	salary	age	city
102	Priya Reddy	priya.reddy@gmail.com	HR	HR Executive	45000.00	26	Vijayawada
103	Arjun Kumar	arjun.kumar@gmail.com	Finance	Financial Analyst	55000.00	28	Bangalore
104	Sneha Patel	sneha.patel@gmail.com	Marketing	Marketing Executive	50000.00	25	Chennai
105	Vikram Singh	vikram.singh@gmail.com	Sales	Sales Executive	45000.00	27	Hyderabad
107	Karthik Rao	karthik.rao@gmail.com	Finance	Accountant	50000.00	30	Vijayawada
108	Divya Nair	divya.nair@gmail.com	HR	Recruiter	45000.00	29	Bangalore
109	Rohan Mehta	rohan.mehta@gmail.com	Marketing	Content Strategist	60000.00	26	Chennai
110	Meera Iyer	meera.iyer@gmail.com	Sales	Business Development Executive	55000.00	24	Pune
112	Kavya Singh	kavya.singh@gmail.com	HR	HR Manager	70000.00	35	Vijayawada
113	Aman Gupta	aman.gupta@gmail.com	Finance	Senior Accountant	65000.00	33	Bangalore
114	Pooja Das	pooja.das@gmail.com	Marketing	SEO Specialist	50000.00	27	Chennai
115	Naveen Kumar	naveen.kumar@gmail.com	Sales	Sales Manager	70000.00	34	Pune

Stage 6: Special Operators

1. **BETWEEN** : Find employees whose salary is between 60,000 and 65,000.

   # BETWEEN is used to find values within a specific range.

SELECT * FROM employees WHERE salary BETWEEN 60000 AND 65000;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	email	department	job_role	salary	age	city
101	Rahul Sharma	rahul.sharma@gmail.com	IT	Software Developer	60000.00	24	Hyderabad
109	Rohan Mehta	rohan.mehta@gmail.com	Marketing	Content Strategist	60000.00	26	Chennai
111	Suresh Babu	suresh.babu@gmail.com	IT	Data Analyst	65000.00	31	Hyderabad
113	Aman Gupta	aman.gupta@gmail.com	Finance	Senior Accountant	65000.00	33	Bangalore
120	Neha Joshi	neha.joshi@gmail.com	IT	UI UX Designer	60000.00	28	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

2. **IN** : Find employees belonging to selected departments such as Hyd,Vij,Banglore.

   # IN is used to check for multiple selected values.

SELECT * FROM employees WHERE city IN ('Hyderabad', 'Vijayawada', 'Bangalore');

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	email	department	job_role	salary	age	city
101	Rahul Sharma	rahul.sharma@gmail.com	IT	Software Developer	60000.00	24	Hyderabad
102	Priya Reddy	priya.reddy@gmail.com	HR	HR Executive	45000.00	26	Vijayawada
103	Arjun Kumar	arjun.kumar@gmail.com	Finance	Financial Analyst	55000.00	28	Bangalore
105	Vikram Singh	vikram.singh@gmail.com	Sales	Sales Executive	45000.00	27	Hyderabad
107	Karthik Rao	karthik.rao@gmail.com	Finance	Accountant	50000.00	30	Vijayawada
108	Divya Nair	divya.nair@gmail.com	HR	Recruiter	45000.00	29	Bangalore
111	Suresh Babu	suresh.babu@gmail.com	IT	Data Analyst	65000.00	31	Hyderabad
112	Kavya Singh	kavya.singh@gmail.com	HR	HR Manager	70000.00	35	Vijayawada
113	Aman Gupta	aman.gupta@gmail.com	Finance	Senior Accountant	65000.00	33	Bangalore
116	Lakshmi Devi	lakshmi.devi@gmail.com	IT	Project Manager	80000.00	36	Hyderabad
117	Manoj Reddy	manoj.reddy@gmail.com	Finance	Finance Manager	80000.00	38	Vijayawada
118	Swathi Krishna	swathi.krishna@gmail.com	Marketing	Graphic Designer	55000.00	25	Bangalore
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. **NOT IN** : Find employees who are not from IT or HR departments.

   # NOT IN is used to exclude multiple values from the result.

SELECT * FROM employees WHERE department NOT IN ('IT', 'HR');

emp_id	emp_name	email	department	job_role	salary	age	city
103	Arjun Kumar	arjun.kumar@gmail.com	Finance	Financial Analyst	55000.00	28	Bangalore
104	Sneha Patel	sneha.patel@gmail.com	Marketing	Marketing Executive	50000.00	25	Chennai
105	Vikram Singh	vikram.singh@gmail.com	Sales	Sales Executive	45000.00	27	Hyderabad
107	Karthik Rao	karthik.rao@gmail.com	Finance	Accountant	50000.00	30	Vijayawada
109	Rohan Mehta	rohan.mehta@gmail.com	Marketing	Content Strategist	60000.00	26	Chennai
110	Meera Iyer	meera.iyer@gmail.com	Sales	Business Development Executive	55000.00	24	Pune
113	Aman Gupta	aman.gupta@gmail.com	Finance	Senior Accountant	65000.00	33	Bangalore
114	Pooja Das	pooja.das@gmail.com	Marketing	SEO Specialist	50000.00	27	Chennai
115	Naveen Kumar	naveen.kumar@gmail.com	Sales	Sales Manager	70000.00	34	Pune
117	Manoj Reddy	manoj.reddy@gmail.com	Finance	Finance Manager	80000.00	38	Vijayawada
118	Swathi Krishna	swathi.krishna@gmail.com	Marketing	Graphic Designer	55000.00	25	Bangalore
119	Ajay Varma	ajay.varma@gmail.com	Sales	Sales Associate	40000.00	22	Chennai
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

4. **LIKE :** LIKE is used to search for patterns.

Employee names starting with 'A'

LIKE is used to search for a particular pattern in text, such as employee names starting with a specific letter.

SELECT *FROM employees WHERE emp_name LIKE 'A%';

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	email	department	job_role	salary	age	city
103	Arjun Kumar	arjun.kumar@gmail.com	Finance	Financial Analyst	55000.00	28	Bangalore
106	Anjali Verma	anjali.verma@gmail.com	IT	Software Tester	55000.00	23	Pune
113	Aman Gupta	aman.gupta@gmail.com	Finance	Senior Accountant	65000.00	33	Bangalore
119	Ajay Varma	ajay.varma@gmail.com	Sales	Sales Associate	40000.00	22	Chennai
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

4. **IS NULL** : Find employees whose city value is NULL.

IS NULL is used to find records where a value is missing,

SELECT * FROM employees WHERE city IS NULL;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap C

emp_id	emp_name	email	department	job_role	salary	age	city
120	Neha Joshi	neha.joshi@gmail.com	IT	UI UX Designer	60000.00	28	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

5. **IS NOT NULL :** Find employees whose city information is available.
   # while IS NOT NULL is used to find records where a value is available

SELECT * FROM employees WHERE city IS NOT NULL;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	email	department	job_role	salary	age	city
101	Rahul Sharma	rahul.sharma@gmail.com	IT	Software Developer	60000.00	24	Hyderabad
102	Priya Reddy	priya.reddy@gmail.com	HR	HR Executive	45000.00	26	Vijayawada
103	Arjun Kumar	arjun.kumar@gmail.com	Finance	Financial Analyst	55000.00	28	Bangalore
104	Sneha Patel	sneha.patel@gmail.com	Marketing	Marketing Executive	50000.00	25	Chennai
105	Vikram Singh	vikram.singh@gmail.com	Sales	Sales Executive	45000.00	27	Hyderabad
106	Anjali Verma	anjali.verma@gmail.com	IT	Software Tester	55000.00	23	Pune
107	Karthik Rao	karthik.rao@gmail.com	Finance	Accountant	50000.00	30	Vijayawada
108	Divya Nair	divya.nair@gmail.com	HR	Recruiter	45000.00	29	Bangalore
109	Rohan Mehta	rohan.mehta@gmail.com	Marketing	Content Strategist	60000.00	26	Chennai
110	Meera Iyer	meera.iyer@gmail.com	Sales	Business Development Executive	55000.00	24	Pune
111	Suresh Babu	suresh.babu@gmail.com	IT	Data Analyst	65000.00	31	Hyderabad
112	Kavya Singh	kavya.singh@gmail.com	HR	HR Manager	70000.00	35	Vijayawada
113	Aman Gupta	aman.gupta@gmail.com	Finance	Senior Accountant	65000.00	33	Bangalore
114	Pooja Das	pooja.das@gmail.com	Marketing	SEO Specialist	50000.00	27	Chennai
115	Naveen Kumar	naveen.kumar@gmail.com	Sales	Sales Manager	70000.00	34	Pune
116	Lakshmi Devi	lakshmi.devi@gmail.com	IT	Project Manager	80000.00	36	Hyderabad
117	Manoj Reddy	manoj.reddy@gmail.com	Finance	Finance Manager	80000.00	38	Vijayawada
118	Swathi Krishna	swathi.krishna@gmail.com	Marketing	Graphic Designer	55000.00	25	Bangalore
119	Ajay Varma	ajay.varma@gmail.com	Sales	Sales Associate	40000.00	22	Chennai
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

In this stage, I learned how to use special operators in SQL , to filter records in different ways.

These operators make it easier to search and filter the required employee records from the table.

Stage 7: ORDER BY

ORDER BY is used to sort records in ascending or descending order.

1. Sort employees by salary in ascending order
   # ASC is used to sort the records from smaller to larger values -- or in alphabetical order.

SELECT * FROM employees ORDER BY salary ASC;

MySQL Practical Assignment – Employee Data Analysis

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	email	department	job_role	salary	age	city
119	Ajay Varma	ajay.varma@gmail.com	Sales	Sales Associate	40000.00	22	Chennai
102	Priya Reddy	priya.reddy@gmail.com	HR	HR Executive	45000.00	26	Vijayawada
105	Vikram Singh	vikram.singh@gmail.com	Sales	Sales Executive	45000.00	27	Hyderabad
108	Divya Nair	divya.nair@gmail.com	HR	Recruiter	45000.00	29	Bangalore
104	Sneha Patel	sneha.patel@gmail.com	Marketing	Marketing Executive	50000.00	25	Chennai
107	Karthik Rao	karthik.rao@gmail.com	Finance	Accountant	50000.00	30	Vijayawada
114	Pooja Das	pooja.das@gmail.com	Marketing	SEO Specialist	50000.00	27	Chennai
103	Arjun Kumar	arjun.kumar@gmail.com	Finance	Financial Analyst	55000.00	28	Bangalore
106	Anjali Verma	anjali.verma@gmail.com	IT	Software Tester	55000.00	23	Pune
110	Meera Iyer	meera.iyer@gmail.com	Sales	Business Development Executive	55000.00	24	Pune
118	Swathi Kris...	swathi.krishna@gmail.com	Marketing	Graphic Designer	55000.00	25	Bangalore
101	Rahul Sharma	rahul.sharma@gmail.com	IT	Software Developer	60000.00	24	Hyderabad
109	Rohan Mehta	rohan.mehta@gmail.com	Marketing	Content Strategist	60000.00	26	Chennai
120	Neha Joshi	neha.joshi@gmail.com	IT	UI UX Designer	60000.00	28	NULL
111	Suresh Babu	suresh.babu@gmail.com	IT	Data Analyst	65000.00	31	Hyderabad
113	Aman Gupta	aman.gupta@gmail.com	Finance	Senior Accountant	65000.00	33	Bangalore
112	Kavya Singh	kavya.singh@gmail.com	HR	HR Manager	70000.00	35	Vijayawada
115	Naveen Ku...	naveen.kumar@gmail.com	Sales	Sales Manager	70000.00	34	Pune
116	Lakshmi Devi	lakshmi.devi@gmail.com	IT	Project Manager	80000.00	36	Hyderabad
117	Manoj Reddy	manoj.reddy@gmail.com	Finance	Finance Manager	80000.00	38	Vijayawada
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

2. Sort employees by salary in descending order

   # DESC is used to sort the records from larger to smaller values -- or in reverse alphabetical order

SELECT * FROM employees ORDER BY salary DESC;

3. Sort employees by age in descending order

SELECT * FROM employees ORDER BY age DESC;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	email	department	job_role	salary	age	city
117	Manoj Reddy	manoj.reddy@gmail.com	Finance	Finance Manager	80000.00	38	Vijayawada
116	Lakshmi Devi	lakshmi.devi@gmail.com	IT	Project Manager	80000.00	36	Hyderabad
112	Kavya Singh	kavya.singh@gmail.com	HR	HR Manager	70000.00	35	Vijayawada
115	Naveen Kumar	naveen.kumar@gmail.com	Sales	Sales Manager	70000.00	34	Pune
113	Aman Gupta	aman.gupta@gmail.com	Finance	Senior Accountant	65000.00	33	Bangalore
111	Suresh Babu	suresh.babu@gmail.com	IT	Data Analyst	65000.00	31	Hyderabad
107	Karthik Rao	karthik.rao@gmail.com	Finance	Accountant	50000.00	30	Vijayawada
108	Divya Nair	divya.nair@gmail.com	HR	Recruiter	45000.00	29	Bangalore
103	Arjun Kumar	arjun.kumar@gmail.com	Finance	Financial Analyst	55000.00	28	Bangalore
120	Neha Joshi	neha.joshi@gmail.com	IT	UI UX Designer	UI UX Designer	28	NULL
105	Vikram Singh	vikram.singh@gmail.com	Sales	Sales Executive	45000.00	27	Hyderabad
114	Pooja Das	pooja.das@gmail.com	Marketing	SEO Specialist	50000.00	27	Chennai
102	Priya Reddy	priya.reddy@gmail.com	HR	HR Executive	45000.00	26	Vijayawada
109	Rohan Mehta	rohan.mehta@gmail.com	Marketing	Content Strategist	60000.00	26	Chennai
104	Sneha Patel	sneha.patel@gmail.com	Marketing	Marketing Executive	50000.00	25	Chennai
118	Swathi Krishna	swathi.krishna@gmail.com	Marketing	Graphic Designer	55000.00	25	Bangalore
101	Rahul Sharma	rahul.sharma@gmail.com	IT	Software Developer	60000.00	24	Hyderabad
110	Meera Iyer	meera.iyer@gmail.com	Sales	Business Develop...	55000.00	24	Pune
106	Anjali Verma	anjali.verma@gmail.com	IT	Software Tester	55000.00	23	Pune
119	Ajay Varma	ajay.varma@gmail.com	Sales	Sales Associate	40000.00	22	Chennai
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

4. Sort employees by department

SELECT * FROM employees ORDER BY department ASC;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	email	department	job_role	salary	age	city
103	Arjun Kumar	arjun.kumar@gmail.com	Finance	Financial Analyst	55000.00	28	Bangalore
107	Karthik Rao	karthik.rao@gmail.com	Finance	Accountant	50000.00	30	Vijayawada
113	Aman Gupta	aman.gupta@gmail.com	Finance	Senior Accountant	65000.00	33	Bangalore
117	Manoj Reddy	manoj.reddy@gmail.com	Finance	Finance Manager	80000.00	38	Vijayawada
102	Priya Reddy	priya.reddy@gmail.com	HR	HR Executive	45000.00	26	Vijayawada
108	Divya Nair	divya.nair@gmail.com	HR	Recruiter	45000.00	29	Bangalore
112	Kavya Singh	kavya.singh@gmail.com	HR	HR Manager	70000.00	35	Vijayawada
101	Rahul Sharma	rahul.sharma@gmail.com	IT	Software Developer	60000.00	24	Hyderabad
106	Anjali Verma	anjali.verma@gmail.com	IT	Software Tester	55000.00	23	Pune
111	Suresh Babu	suresh.babu@gmail.com	IT	Data Analyst	65000.00	31	H Hyderabad
116	Lakshmi Devi	lakshmi.devi@gmail.com	IT	Project Manager	80000.00	36	Hyderabad
120	Neha Joshi	neha.joshi@gmail.com	IT	UI UX Designer	60000.00	28	NULL
104	Sneha Patel	sneha.patel@gmail.com	Marketing	Marketing Executive	50000.00	25	Chennai
109	Rohan Mehta	rohan.mehta@gmail.com	Marketing	Content Strategist	60000.00	26	Chennai
114	Pooja Das	pooja.das@gmail.com	Marketing	SEO Specialist	50000.00	27	Chennai
118	Swathi Krishna	swathi.krishna@gmail.com	Marketing	Graphic Designer	55000.00	25	Bangalore
105	Vikram Singh	vikram.singh@gmail.com	Sales	Sales Executive	45000.00	27	Hyderabad
110	Meera Iyer	meera.iyer@gmail.com	Sales	Business Developm...	55000.00	24	Pune
115	Naveen Kumar	naveen.kumar@gmail.com	Sales	Sales Manager	70000.00	34	Pune
119	Ajay Varma	ajay.varma@gmail.com	Sales	Sales Associate	40000.00	22	Chennai
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

** Sort employees by multiple columns

For example, first sort by department in ascending order, then employees with the same department by salary in descending order.

SELECT * FROM employees ORDER BY department ASC, salary DESC;

In this stage, I learned how to use the ORDER BY clause to arrange employee records in a particular order. I also learned that ORDER BY can be used with different columns such as salary, age, and department. We can also sort using multiple columns.

Stage 8: Aggregate Functions

Aggregate functions perform calculations on multiple rows and return a single result.

1. COUNT() – Total Number of Employees

   # This counts all employee records in the table.

   SELECT COUNT(*) AS total_employees FROM employees;

   Result Grid | Fi

total_employees
20

2. SUM() – Total Salary Paid

   # This calculates the total of all employee salaries.

   SELECT SUM(salary) AS total_salary FROM employees;

   Result Grid | F

total_salary
1155000.00

---

3. AVG() – Average Salary

   # This calculates the average salary of all employees.

   SELECT AVG(salary) AS average_salary FROM employees;

	average_salary
▶	57750.000000

4. MIN() – Lowest Salary

   # This finds the lowest salary in the table.

   SELECT MIN(salary) AS minimum_salary FROM employees;

	minimum_salary
▶	40000.00

5. MAX() – Highest Salary

   # This finds the highest salary in the table.

   SELECT MAX(salary) AS maximum_salary FROM employees;

	maximum_salary
▶	80000.00

Stage 9: GROUP BY

GROUP BY is used to group rows that have the same value in a particular column. Then aggregate functions like COUNT(), AVG(), MIN(), MAX(), and SUM() can be applied to each group.

Department-wise analysis:

- Count employees in each department.
- Calculate average salary for each department.
- Find minimum salary in each department.
- Find maximum salary in each department.
- Calculate total salary by department.

```
SELECT  department,
   COUNT(*) AS total_employees,
   AVG(salary) AS average_salary,
   MIN(salary) AS minimum_salary,
   MAX(salary) AS maximum_salary,
   SUM(salary) AS total_salary
```

FROM employees GROUP BY department;

department	total_employees	average_salary	minimum_salary	maximum_salary	total_salary
IT	5	64000.000000	55000.00	80000.00	320000.00
HR	3	53333.333333	45000.00	70000.00	160000.00
Finance	4	62500.000000	50000.00	80000.00	250000.00
Marketing	4	53750.000000	50000.00	60000.00	215000.00
Sales	4	52500.000000	40000.00	70000.00	210000 210000.00

City-wise Analysis

1. **Count employees in each city**

   # GROUP BY will also show employees whose city is NULL as a separate group.

SELECT city, COUNT(*) AS total_employees FROM employees GROUP BY city;

city	total_employees
Hyderabad	4
Vijayawada	4
Bangalore	4
Chennai	4
Pune	3
NULL	1

2. Calculate average salary by city

SELECT city, AVG(salary) AS average_salary FROM employees GROUP BY city;

city	average_salary
Hyderabad	62500.000000
Vijayawada	61250.000000
Bangalore	55000.000000
Chennai	50000.000000
Pune	60000.000000
NULL	60000.000000

3. Find maximum salary by city

SELECT city, MAX(salary) AS maximum_salary FROM employees GROUP BY city;

city	maximum_salary
Hyderabad	80000.00
Vijayawada	80000.00
Bangalore	65000.00
Chennai	60000.00
Pune	70000.00
NULL	60000.00

Stage 10: HAVING

HAVING is used to filter **grouped results**, usually after using GROUP BY.

1. Find departments having more than a specified number of employees

SELECT department, COUNT(*) AS total_employees

FROM employees GROUP BY department HAVING COUNT(*) > 3;

Result Grid | Filter Rows:

department	total_employees
IT	5
Finance	4
Marketing	4
Sales	4

2. Find departments with an average salary above a specified amount

SELECT department, AVG(salary) AS average_salary

FROM employees GROUP BY department HAVING AVG(salary) > 55000;

Result Grid | Filter Rows:

department	average_salary
IT	64000.000000
Finance	62500.000000

3. Find cities having more than a specified number of employees

SELECT city, COUNT(*) AS total_employees FROM employees GROUP BY city HAVING COUNT(*) > 2;

city	total_employees
Hyderabad	4
Vijayawada	4
Bangalore	4
Chennai	4
Pune	3

4. Find departments where the maximum salary is above a specified amount

SELECT department, MAX(salary) AS maximum_salary

FROM employees GROUP BY department HAVING MAX(salary) > 70000;

department	maximum_salary
IT	80000.00
Finance	80000.00

Stage 11: Challenge Questions

1. **Find the second-highest salary**
   # This query first finds the highest salary. Then it finds the maximum salary that is less than the highest salary.

SELECT MAX(salary) AS second_highest_salary

FROM employees WHERE salary < (

SELECT MAX(salary) FROM employees);

Result Grid | Filter Rows:

second_highest_salary
70000.00

2. **Find employees earning above the average salary**
   # The subquery calculates the average salary of all employees. Then ,the main query compares each employee's salary with the average.

SELECT *

FROM employees WHERE salary > (

SELECT AVG(salary) FROM employees);

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	email	department	job_role	salary	age	city
101	Rahul Sharma	rahul.sharma@gmail.com	IT	Software Developer	60000.00	24	Hyderabad
109	Rohan Mehta	rohan.mehta@gmail.com	Marketing	Content Strategist	60000.00	26	Chennai
111	Suresh Babu	suresh.babu@gmail.com	IT	Data Analyst	65000.00	31	Hyderabad
112	Kavya Singh	kavya.singh@gmail.com	HR	HR Manager	70000.00	35	Vijayawada
113	Aman Gupta	aman.gupta@gmail.com	Finance	Senior Accountant	65000.00	33	Bangalore
115	Naveen Kumar	naveen.kumar@gmail.com	Sales	Sales Manager	70000.00	34	Pune
116	Lakshmi Devi	lakshmi.devi@gmail.com	IT	Project Manager	80000.00	36	Hyderabad
117	Manoj Reddy	manoj.reddy@gmail.com	Finance	Finance Manager	80000.00	38	Vijayawada
120	Neha Joshi	neha.joshi@gmail.com	IT	UI UX Designer	60000.00	28	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. **Find the employee(s) with the highest salary**
   # The subquery finds the highest salary using MAX(). The main query finds the employees whose salary having highest salary.

SELECT * FROM employees WHERE salary = (SELECT MAX(salary) FROM employees);

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	email	department	job_role	salary	age	city
116	Lakshmi Devi	lakshmi.devi@gmail.com	IT	Project Manager	80000.00	36	Hyderabad
117	Manoj Reddy	manoj.reddy@gmail.com	Finance	Finance Manager	80000.00	38	Vijayawada
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

4. **Find the employee(s) with the lowest salary**
   # The subquery finds the lowest salary using MIN(). -- The main query finds the employees whose salary is lowest.

SELECT * FROM employees WHERE salary = (

SELECT MIN(salary) FROM employees);

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Conte

emp_id	emp_name	email	department	job_role	salary	age	city
119	Ajay Varma	ajay.varma@gmail.com	Sales	Sales Associate	40000.00	22	Chennai
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

5. **Find the department with the highest average salary**

First, the query calculates the average salary for each department. Then the departments are arranged from highest average salary to lowest. LIMIT 1 displays the department with the highest average salary.

SELECT department, AVG(salary) AS average_salary

FROM employees GROUP BY department

ORDER BY average_salary DESC

LIMIT 1;

Result Grid | Filter Rows:

department	average_salary
IT	64000.000000

6. **Find the department with the highest number of employees**
   # COUNT() is used to find the number of employees in each department. The departments are then sorted from highest count to lowest count.

SELECT department, COUNT(*) AS total_employees

FROM employees GROUP BY department

ORDER BY total_employees DESC

LIMIT 1;

Result Grid | Filter Rows:

department	total_employees
IT	5

7. Find employees who earn more than the average salary of their department

SELECT * FROM employees e WHERE salary > (

SELECT AVG(salary) FROM employees

WHERE department = e.department

);

It checks that employees department.

Calculates the average salary of that department.

Displays the employee only if their salary is greater than that average.

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	email	department	job_role	salary	age	city
109	Rohan Mehta	rohan.mehta@gmail.com	Marketing	Content Strategist	60000.00	26	Chennai
110	Meera Iyer	meera.iyer@gmail.com	Sales	Business Development Executive	55000.00	24	Pune
111	Suresh Babu	suresh.babu@gmail.com	IT	Data Analyst	65000.00	31	Hyderabad
112	Kavya Singh	kavya.singh@gmail.com	HR	HR Manager	70000.00	35	Vijayawada
113	Aman Gupta	aman.gupta@gmail.com	Finance	Senior Accountant	65000.00	33	Bangalore
115	Naveen Kumar	naveen.kumar@gmail.com	Sales	Sales Manager	70000.00	34	Pune
116	Lakshmi Devi	lakshmi.devi@gmail.com	IT	Project Manager	80000.00	36	Hyderabad
117	Manoj Reddy	manoj.reddy@gmail.com	Finance	Finance Manager	80000.00	38	Vijayawada
118	Swathi Krishna	swathi.krishna@gmail.com	Marketing	Graphic Designer	55000.00	25	Bangalore
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

8. Find the second-highest salary in each department

The inner query finds highest salary in that department.

WHERE salary < highest salary removes the highest salary.

MAX(salary) finds the highest remaining salary.GROUP BY department gives the result for each department.

```
SELECT department, MAX(salary) AS second_highest_salary
FROM employees e1
WHERE salary < (
   SELECT MAX(salary) FROM employees e2 WHERE e2.department =
e1.department)
GROUP BY department;
```

Or

`DENSE_RANK()` is useful because if multiple employees have the same salary, they receive the same rank.

```
SELECT department, salary AS second_highest_salary
FROM (
   SELECT department,salary,
  DENSE_RANK() OVER (
 PARTITION BY department ORDER BY salary DESC
) AS salary_rank
   FROM employees) AS ranked_salaries
WHERE salary_rank = 2;
```

Result Grid | Filter Rows:

department	second_highest_salary
IT	65000.00
HR	45000.00
Finance	65000.00
Marketing	55000.00
Sales	55000.00

9. Find cities with the highest number of employees

First, the inner query counts the employees in each city. The outer query finds the highest employee count.HAVING is used to display the city or cities that have this highest number of employees.

```
SELECT city, COUNT(*) AS total_employees
FROM employees GROUP BY city
HAVING COUNT(*) = (
  SELECT MAX(employee_count)
      FROM (
    SELECT COUNT(*) AS employee_count
    FROM employees
    GROUP BY city
  ) AS city_counts);
```

Result Grid | Filter Rows:

city	total_employees
Hyderabad	4
Vijayawada	4
Bangalore	4
Chennai	4

10. Find departments where the total salary is greater than a specified amount

SUM() calculates the total salary for each department. GROUP BY is used to calculate the total separately for each department. HAVING filters the departments based on their total salary.

```
SELECT department, SUM(salary) AS total_salary
FROM employees GROUP BY department
HAVING SUM(salary) > 250000;
```

Result Grid | Filter Rows:

department	total_salary
IT	320000.00

Stage 12: Final Deliverables
To Verify

Query 1: Verify Table Structure

DESC employees;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
emp_id	int	NO	PRI	NULL	
emp_name	varchar(100)	NO		NULL	
email	varchar(100)	NO	UNI	NULL	
department	varchar(50)	NO		NULL	
job_role	varchar(50)	NO		NULL	
salary	decimal(10,2)	YES		NULL	
age	int	YES		NULL	
city	varchar(50)	YES		Vijayawada	

Query 2 : By Display all employee records

SELECT * FROM employees;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	email	department	job_role	salary	age	city
101	Rahul Sharma	rahul.sharma@gmail.com	IT	Software Developer	60000.00	24	Hyderabad
102	Priya Reddy	priya.reddy@gmail.com	HR	HR Executive	45000.00	26	Vijayawada
103	Arjun Kumar	arjun.kumar@gmail.com	Finance	Financial Analyst	55000.00	28	Bangalore
104	Sneha Patel	sneha.patel@gmail.com	Marketing	Marketing Executive	50000.00	25	Chennai
105	Vikram Singh	vikram.singh@gmail.com	Sales	Sales Executive	45000.00	27	Hyderabad
106	Anjali Verma	anjali.verma@gmail.com	IT	Software Tester	55000.00	23	Pune
107	Karthik Rao	karthik.rao@gmail.com	Finance	Accountant	50000.00	30	Vijayawada
108	Divya Nair	divya.nair@gmail.com	HR	Recruiter	45000.00	29	Bangalore
109	Rohan Mehta	rohan.mehta@gmail.com	Marketing	Content Strategist	60000.00	26	Chennai
110	Meera Iyer	meera.iyer@gmail.com	Sales	Business Development Executive	55000.00	24	Pune
111	Suresh Babu	suresh.babu@gmail.com	IT	Data Analyst	65000.00	31	Hyderabad
112	Kavya Singh	kavya.singh@gmail.com	HR	HR Manager	70000.00	35	Vijayawada
113	Aman Gupta	aman.gupta@gmail.com	Finance	Senior Accountant	65000.00	33	Bangalore
114	Pooja Das	pooja.das@gmail.com	Marketing	SEO Specialist	50000.00	27	Chennai
115	Naveen Kumar	naveen.kumar@gmail.com	Sales	Sales Manager	70000.00	34	Pune
116	Lakshmi Devi	lakshmi.devi@gmail.com	IT	Project Manager	80000.00	36	Hyderabad
117	Manoj Reddy	manoj.reddy@gmail.com	Finance	Finance Manager	80000.00	38	Vijayawada
118	Swathi Krishna	swathi.krishna@gmail.com	Marketing	Graphic Designer	55000.00	25	Bangalore
119	Ajay Varma	ajay.varma@gmail.com	Sales	Sales Associate	40000.00	22	Chennai
120	Neha Joshi	neha.joshi@gmail.com	IT	UI UX Designer	60000.00	28	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Query 3: Verify Total Employees

SELECT COUNT(*) AS total_employees FROM employees;

Result Grid | Filter

	total_employees
▶	20